Beyond cooperative pairs: two whole-brain papers of 2026

Computational Neuroscience 2026/27 · Lesson 2, first part

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Thursday 5 November 2026, room B

Luppi et al., *Nat Neurosci* 29:915 (2026) · Santoro et al., *Nat Commun* 17:9207 (2026)

Outline

Thirty minutes, two papers

Opening	What the model of Lesson 1 cannot do	3 min
Paper 1	Luppi et al.: competitive interactions in mammalian connectomes	14 min
Paper 2	Santoro et al.: higher-order interactions beyond pairs	10 min
Discussion	One synthesis, four questions	3 min

Both papers are open access

From Lesson 1

What the cooperative pairwise model misses

The Hopf model of Lesson 1 couples regions with one positive weight per fibre, $GC_{ij} \geq 0$, symmetric.

1. Its FC fits the measured FC at about 0.66, with one free parameter.
2. With all couplings ≥ 0 and symmetric, and the same frequency in every region, every partial correlation is ≥ 0: every triplet has $\Omega \geq 0$, none is synergistic. Negative or asymmetric couplings, or different frequencies, can remove this constraint.
3. With symmetric C and equal frequencies, lagged covariances are symmetric: no direction.

Two ways forward

Change the model: let couplings take either sign (paper 1).

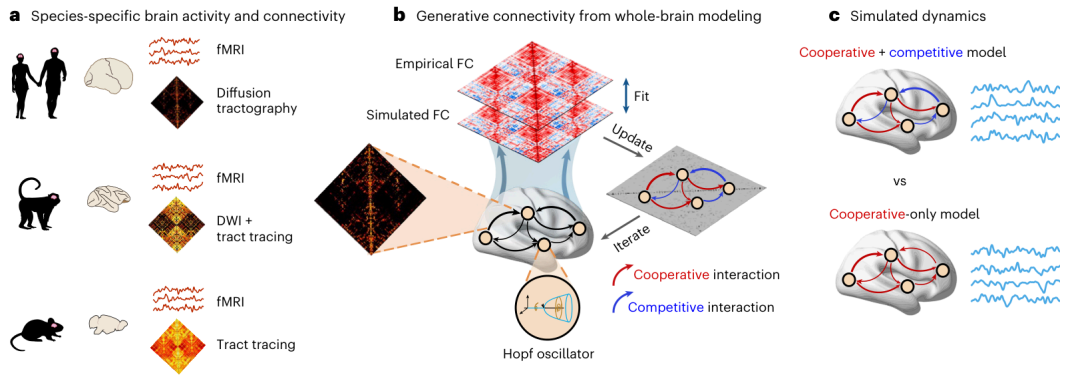
Change the observables: measure dependence beyond pairs at one instant, among three regions or across time (paper 2).

Competitive interactions shape mammalian brain network dynamics and computation

Luppi, Sanz Perl, Vohryzek, ..., Misic, Deco & Kringelbach · *Nature Neuroscience* 29:915–933 (2026)

doi:10.1038/s41593-026-02205-3 · *CC BY 4.0*

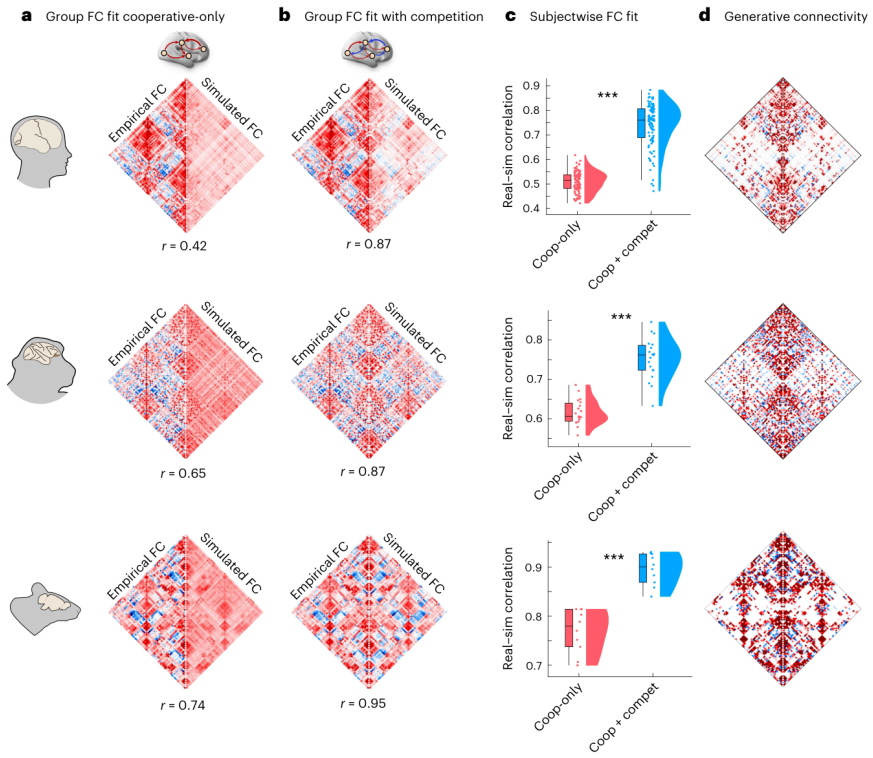
A whole-brain model in three species, with signed couplings



1. Hopf nodes just below the bifurcation, as in Lesson 1, but each region with its own frequency, coupled by species-specific SC: human (100 subjects, tractography), macaque (19 scans of 10 animals, tractography and tract tracing), mouse (10 animals, tract tracing).
2. **Generative connectivity** (GC): every existing SC weight is updated, subject by subject, until the model reproduces the measured FC and the lagged FC, as the GEC of the book.
3. Two versions: **cooperative-only** (all weights ≥ 0) and **cooperative–competitive** (the sign is free). A negative weight from A to B: more activity in A, less in B.

Luppi et al. (2026), Fig. 1. CC BY 4.0

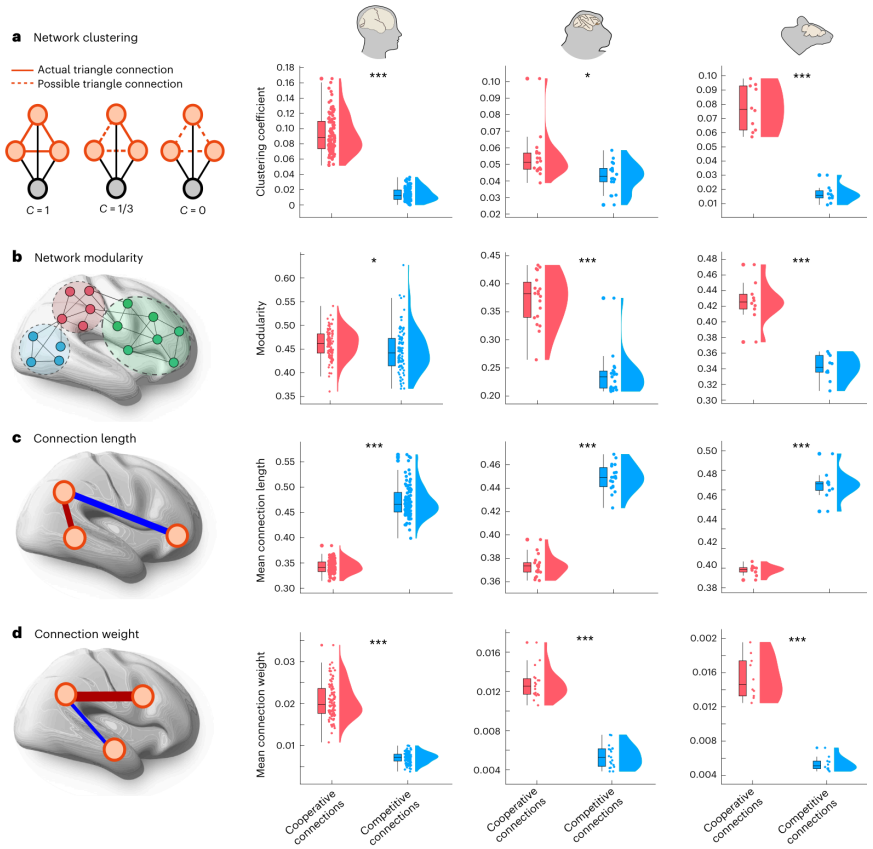
Negative weights appear, and the fit improves



1. Negative weights are allowed, not imposed. The fit makes 25–40% of them negative: human 25 ± 8%, macaque 38 ± 7%, mouse 28 ± 4%.
2. Fit per subject, the correlation of simulated and measured FC: human 0.51 → 0.74, macaque 0.61 → 0.75, mouse 0.70 → 0.89 ($P < 0.001$).
3. No parameter is added: the same weights take a wider range. A cooperative-only model with twice as many connections still fits worse.
4. A cooperative-only model can produce negative FC too, up to about 30% of the pairs: unlike Lesson 1, the frequencies differ between regions, and with different frequencies positive couplings can give negative correlations.

Luppi et al. (2026), Fig. 2. CC BY 4.0

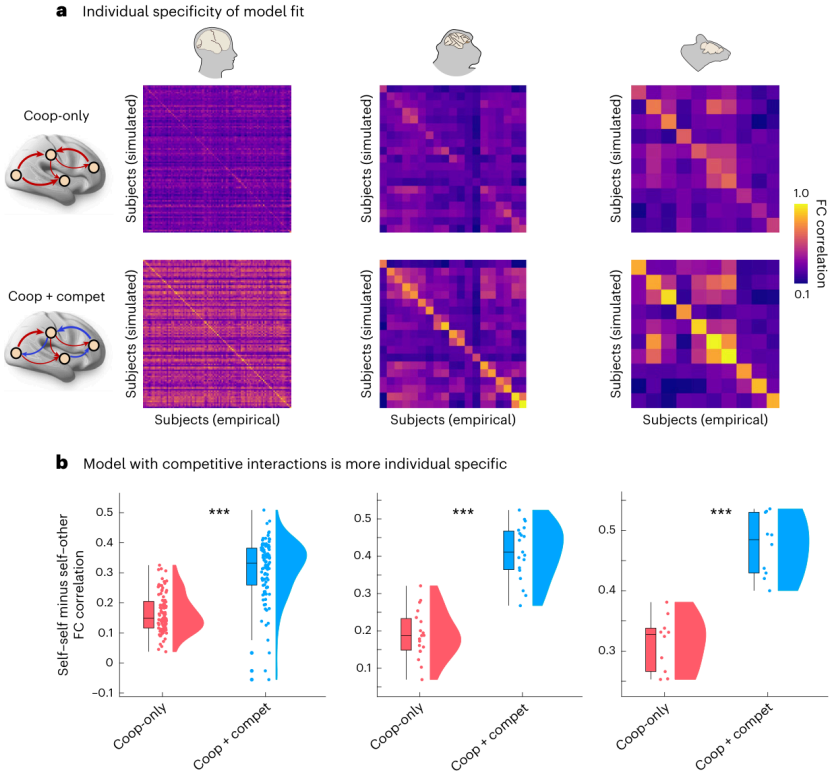
Where the competitive links are



1. **Network.** Competitive connections are less clustered and less modular than cooperative ones, longer in space, and weaker (left).
2. **Biology.** They link regions at opposite ends of biological axes: the ratio of two classes of inhibitory neurons (PV against SST), the myelin hierarchy (T1w/T2w), cell-type composition, gene and receptor expression (their Fig. 4).
3. So cooperation is modular and local; competition is diffuse and long-range, between dissimilar regions.

Luppi et al. (2026), Fig. 3. CC BY 4.0

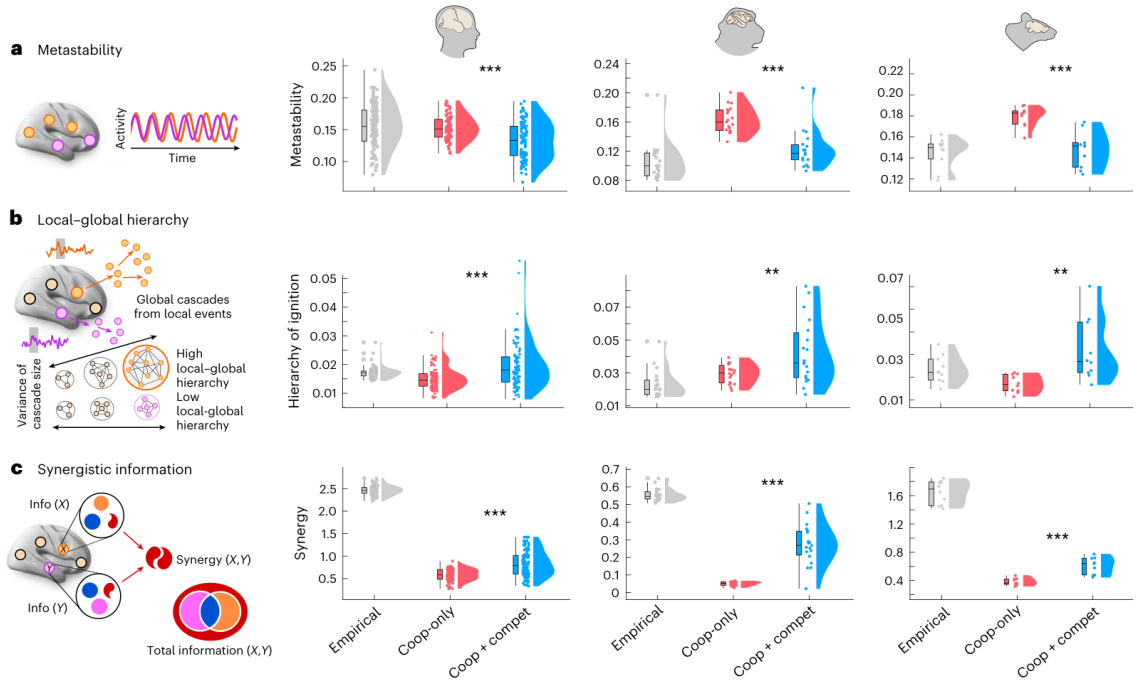
More subject-specific, with checks against overfitting



1. **Differential identifiability**: the similarity of a subject's simulated FC to its own measured FC, minus its mean similarity to other subjects. It is higher with competition.
2. The fitted model reproduces unseen data of the same individual, and fails to fit biologically implausible null data.
3. Interpretation: these checks argue against overfitting alone. They do not show that the weights are anatomical, nor that this mechanism is the only one that fits.

Luppi et al. (2026), Fig. 5. CC BY 4.0

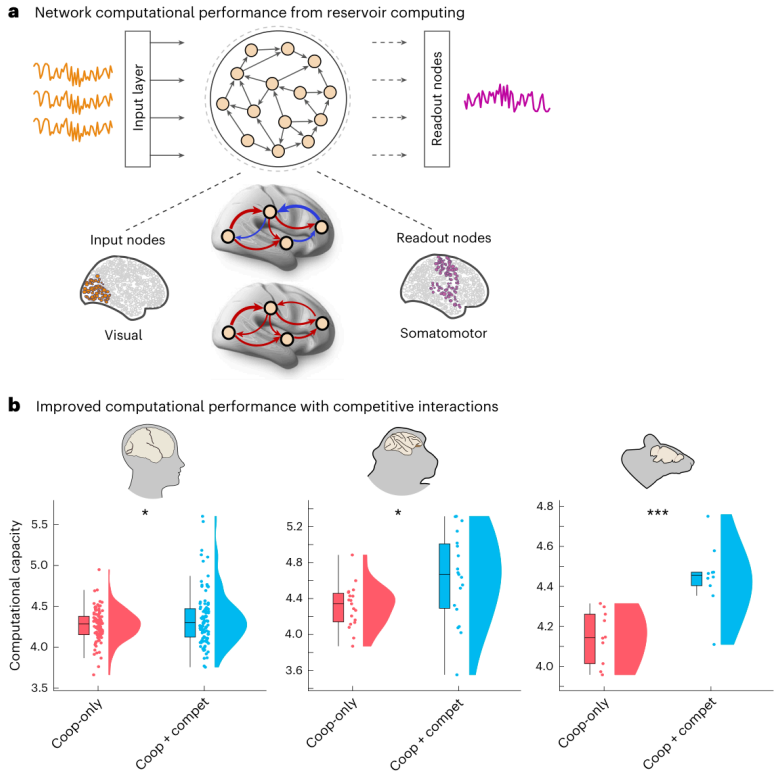
Dynamics that were not fitted



1. **Metastability**, the standard deviation in time of global phase synchrony (the Kuramoto order parameter), and the **local–global hierarchy** of spontaneous events move towards the data.
2. **Synergy in time** (ΦID: information that two regions carry only jointly, in their past and still in their future) increases with competition, though it stays below the data. It is not the **Ω** of triplets of Lessons 0–1.
3. Spontaneous activity resembles more closely 123 meta-analytic maps of cognitive tasks (their Fig. 7).

Luppi et al. (2026), Fig. 6. CC BY 4.0

Better memory in a connectome-wired network



1. **Reservoir computing**: a recurrent network of nonlinear units wired with the generative connectivity. Input enters visual regions; a linear readout from motor regions is trained to reproduce the input of some steps before.
2. Memory capacity is higher with competitive weights in most comparisons, also against cooperative-only networks with twice the connections.

Caveat

A negative weight is a net effect. Inhibition, conduction delays or the haemodynamics could produce it: the model does not say which.

Luppi et al. (2026), Fig. 8. CC BY 4.0

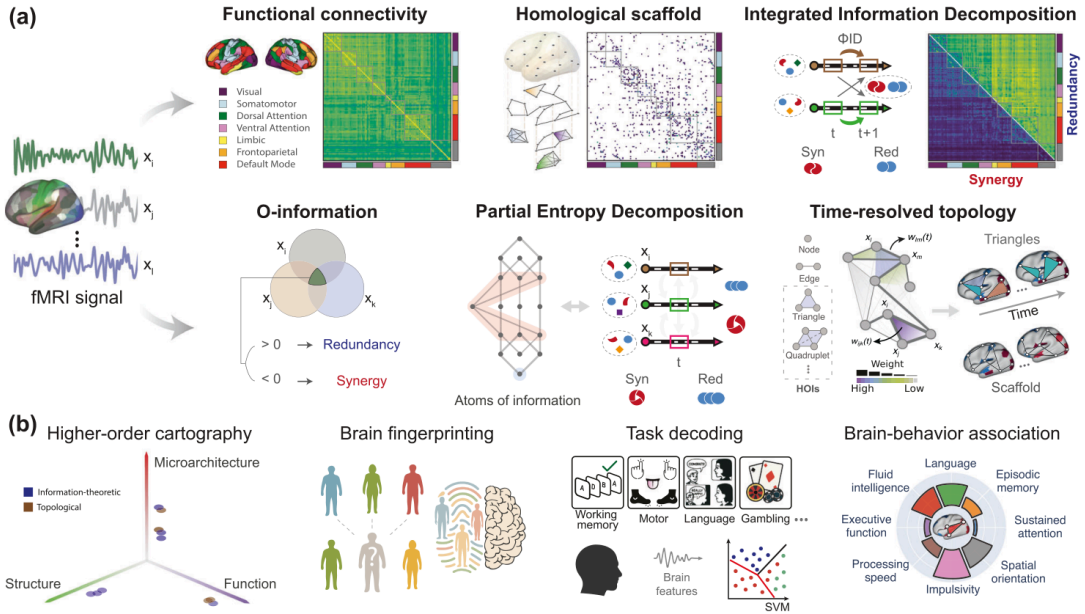
Charting higher-order models of brain function beyond pairwise interactions

Santoro, Neri, Poetto, Orsenigo, Diano, Gatica & Petri · *Nature Communications* 17:9207 (2026)

doi:10.1038/s41467-026-75959-w · *CC BY-NC-ND 4.0*

Paper 2 · Question

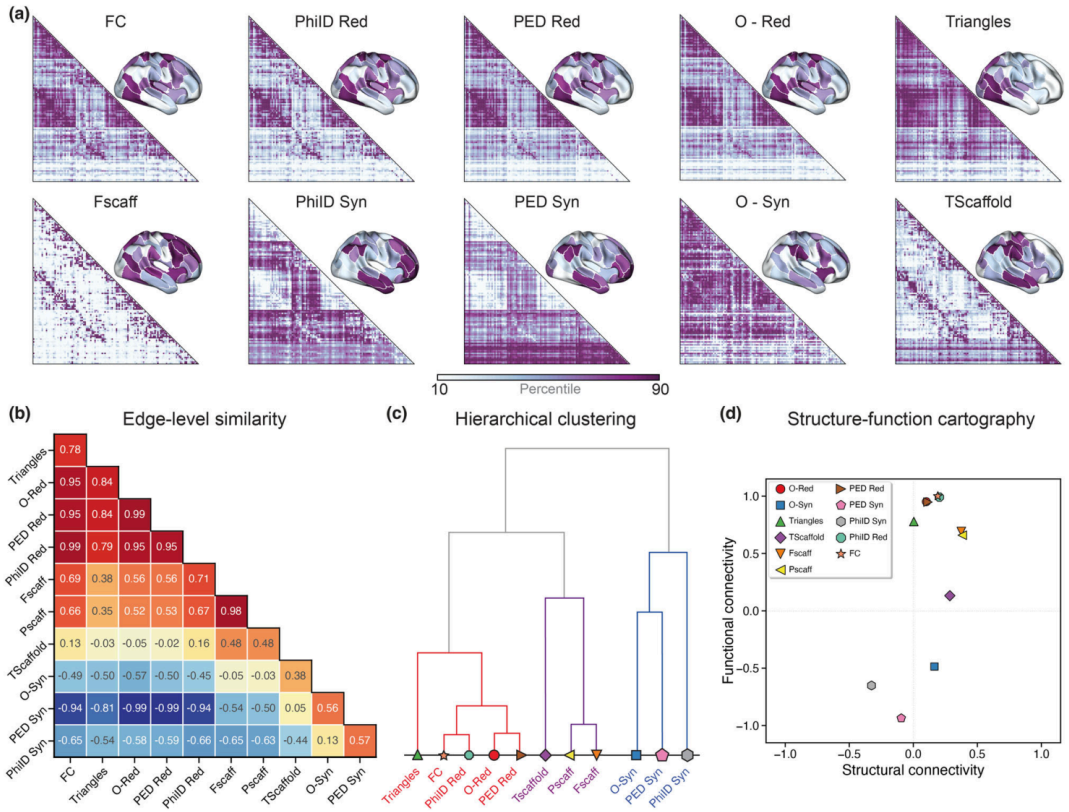
Many higher-order metrics: which say what?



1. **Higher-order interaction** (HOI) metrics. Information: redundancy and synergy from O-information (Lesson 0) and PED on triplets, ΦID on pairs in past and future. Topology: triangles and scaffolds of co-fluctuations, homological scaffolds of the FC.
2. Data: 100 unrelated HCP subjects, rest and seven tasks, 116 regions (100 cortical, 16 subcortical). Every metric is compared through its projection onto regions and pairs; FC is the pairwise baseline.

Santoro et al. (2026), Fig. 1, reproduced without changes. CC BY-NC-ND 4.0

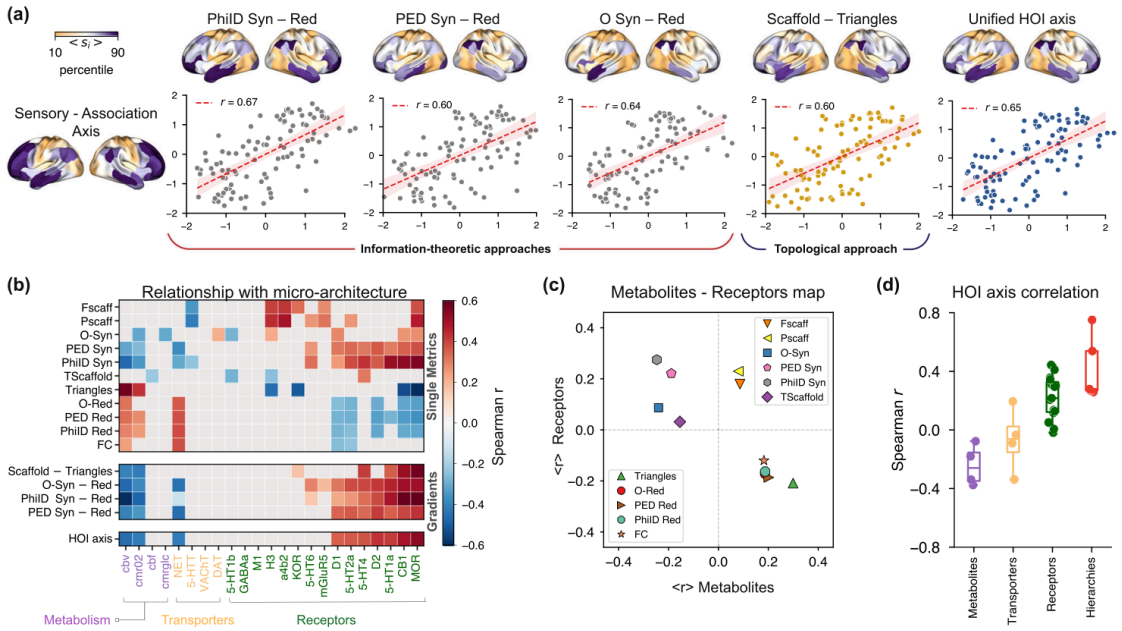
Three families: redundancy, topology, synergy



1. Hierarchical clustering of the edge patterns separates redundancy metrics (FC, the redundancy parts of ΦID, PED and O-information, triangles), topological ones, and synergy metrics.
2. Redundancy weights sensory cortex and pairs within the same network; synergy weights transmodal cortex (frontoparietal, default mode) and pairs between networks.
3. Redundancy metrics follow the SC and the FC; synergy metrics do not.

Santoro et al. (2026), Fig. 2, reproduced without changes. CC BY-NC-ND 4.0

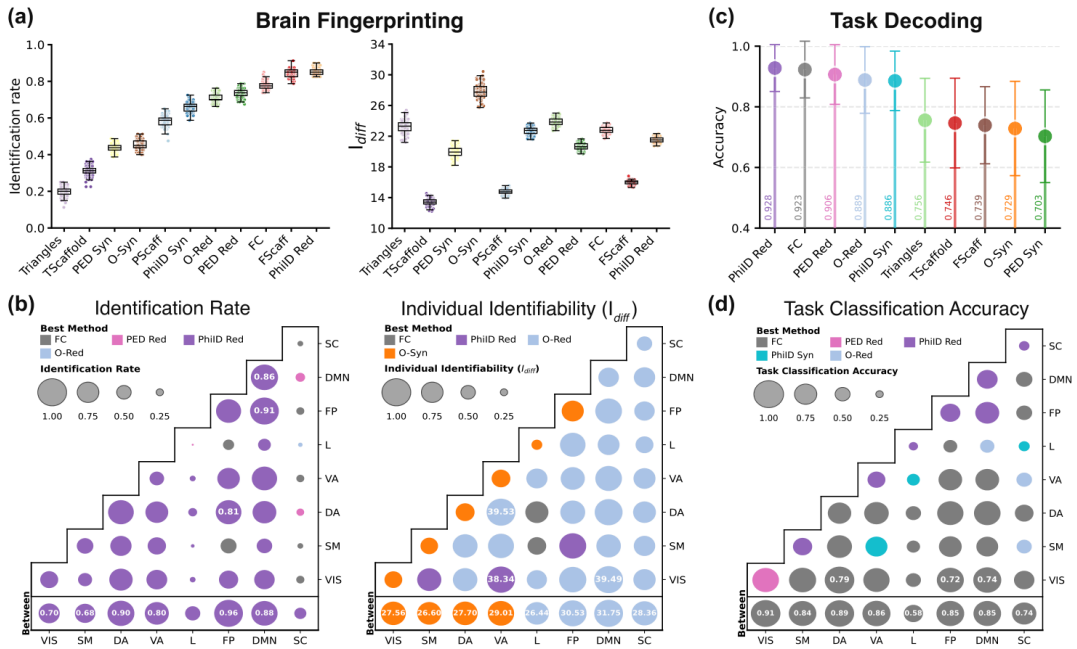
One axis, two molecular signatures



1. For each region, synergy minus redundancy, from ΦID, PED, O-information, and scaffold minus triangles: the four gradients agree ($\rho > 0.8$) and follow the sensorimotor–association axis (Spearman $>$ 0.6).
2. Their mean, a unified HOI axis, correlates positively with receptor maps and negatively with metabolic maps (spatial null models that preserve autocorrelation).

Santoro et al. (2026), Fig. 3, reproduced without changes. CC BY-NC-ND 4.0

Better fingerprints, not better task decoding



1. **Fingerprinting**: identifying a subject across sessions. HOI metrics beat FC in identification rate and identifiability, most in transmodal networks.
2. **Task decoding**: classifying which task is running (linear classifier, leave one subject out). FC stays among the best.
3. **Behaviour**: in a multivariate brain–behaviour analysis, topological descriptors carry the strongest association (their Fig. 6).

Santoro et al. (2026), Fig. 4, reproduced without changes. CC BY-NC-ND 4.0

Discussion

One synthesis

1. **Lesson 1.** Positive, symmetric couplings and equal frequencies: FC fitted at about 0.66, no synergistic triplet, no direction.
2. **Paper 1.** Negative generative couplings, 25–40% of the weights: higher fit (0.74–0.89), more subject-specific models, more synergy in time (ΦID), better memory in a connectome-wired network.
3. **Paper 2.** Higher-order metrics split redundancy (sensory, within networks, metabolism) from synergy (transmodal, between networks, receptors) along the sensorimotor–association axis; they identify individuals better than FC.

Open

Does the competitive model reproduce the higher-order structure of the data? It raises the synergy in time of pairs (ΦID), but not to the level of the data. Triplets at one instant, as in Lessons 0 and 1, are a separate question.

Luppi et al. (2026); Santoro et al. (2026)

Four questions

1. A negative generative weight: synaptic inhibition, conduction delays, or haemodynamics? Which measurement could tell them apart?
2. A cooperative-only model already produces some negative FC. Which feature of the data, then, requires negative couplings: the zero-lag FC or the lagged FC?
3. Why would synergy help to identify a person but not to decode a task?
4. For Gaussian signals, the Ω of a triplet follows from its three correlations (Lesson 0). Which metrics of paper 2 can see beyond the covariance?

The two papers

Luppi AI, Sanz Perl Y, Vohryzek J, Ali H, Mediano PAM, Rosas FE, Milisav F, Suárez LE, Gini S, Gutierrez-Barragan D, Yee Y, Froudist-Walsh S, Gozzi A, Misic B, Deco G, Kringelbach ML (2026). Competitive interactions shape mammalian brain network dynamics and computation. *Nat Neurosci* 29:915–933. doi:10.1038/s41593-026-02205-3. CC BY 4.0.

Santoro A, Neri M, Poetto S, Orsenigo D, Diano M, Gatica M, Petri G (2026). Charting higher-order models of brain function beyond pairwise interactions. *Nat Commun* 17:9207. doi:10.1038/s41467-026-75959-w. CC BY-NC-ND 4.0.

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